

String and math expressions

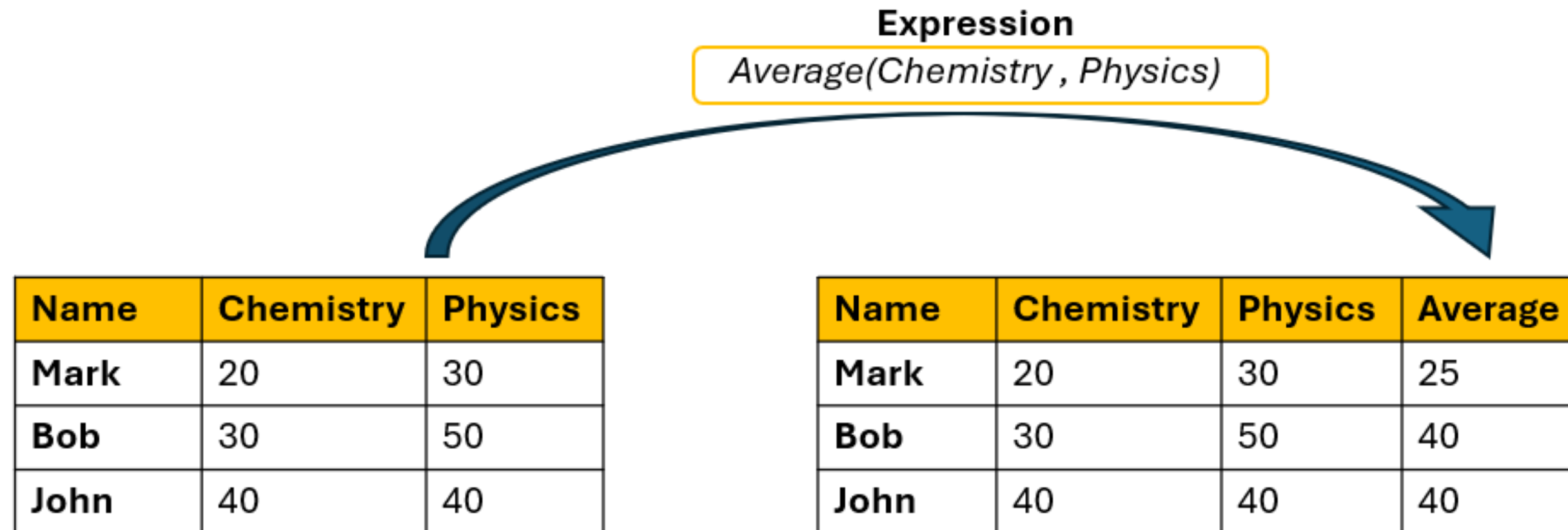
DATA TRANSFORMATION IN KNIME



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Expression-based transformation

- Allows applying custom logic
- Manipulate data with expressions
- Simple arithmetic to complex string manipulations



Formulating an expression

An expression can be formulated by

- Using operators

+ - = * /  Operators

- Using functions

join(),mean(),max()  Functions

- Combination of operators and functions

mean() + max()  Combination

Expression Node

Expression



- Functions for string manipulation
- Math formulas
- Use a range of operators
- Create or replace a new column

String expressions

- Perform simple and complex string manipulations
- Use one or more string functions

Example

```
replace("I loved the m@vie", "@", "o")
```

Output

```
I loved the movie
```

Popular string functions

Name	Expression	Meaning
lowercase, uppercase	lowercase(x) uppercase(x)	Convert the entire string to lower/upper case
capitalize	capitalize(x)	Capitalize first letter of each word

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join	join(separator, x1, x2)	Join 2 or more strings with separator

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join	join(separator, x1, x2)	Join 2 or more strings with separator
replace	replace(x,pattern,replacement)	Replace pattern with replacement
find	find(x,string)	Find the string location index in x
length	length(x)	Length of input string

Math expressions

- Perform mathematical transformation
- Use math operators or functions and formulas

Example

```
max(20, 30, 70+5)
```

Output

```
75
```

Popular math functions

Name	Expression	Meaning
max, min,	max(x), min(x)	Maximum and minimum value

Popular math functions

Name	Expression	Meaning
max, min,	max(x), min(x)	Maximum and minimum value
stddev, average, median	stddev(x), average(x), median(x)	Standard deviation, average and median of value

Popular math functions

Name	Expression	Meaning
max, min,	max(x), min(x)	Maximum and minimum value
stddev, average, median	stddev(x), average(x), median(x)	Standard deviation, average and median of value
log, exp	log(x), exp(x)	Logarithm and exponential of a value
round	round(x, precision)	Round off the value up to precision

Mixed expressions

- Perform string and mathematical transformation.
- Use math and string functions together.

Example

Val1	Val2
20	50

Add a comment column

Val1	Val2	Comment
20	50	The minimum value is 20

Mixed expressions

Calculate minimum value

Syntax

```
min(column1, column2)
```

Expression

```
min($Val1$, $Val2$)
```

Convert it to string format

Syntax

```
string(input)
```

Expression

```
string(min($Val1$, $Val2$))
```

Mixed expressions

Syntax

```
join(separator, string1, string2)
```

Expression

```
join(" ", "The minimum value is", string(min($val1$, $val2$)))
```

Output Comment column

```
The minimum value is 20
```


Let's practice!

DATA TRANSFORMATION IN KNIME

Calculate student scores

DATA TRANSFORMATION IN KNIME



Full Name
Instructor

Let's practice!

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